

## Problem Set 10

### Differentials and Linearization

September 28, 2026

#### Problem 1

(d)

$$\begin{aligned}\lim_{dx \rightarrow 0} \frac{\Delta y - dy}{d^2 x} &= \lim_{dx \rightarrow 0} 7 + dx = 7 \\ f''(2) &= 6 \cdot 2 + 2 = 14 \\ \lim_{dx \rightarrow 0} \frac{\Delta y - dy}{d^2 x} &= \frac{1}{2} f''(2) \\ \Delta y - dy &= \frac{1}{2} f''(2) d^2 x \\ |f(x) - L(x)| &\leq \frac{M}{2} (x - a)^2\end{aligned}$$

#### Problem 2

(a)

$$\begin{aligned}y &= x^2 \sin 2x \\ y' &= 2x \sin 2x + 2x^2 \cos 2x \\ dy &= \boxed{2x(\sin 2x + x \cos 2x) dx}\end{aligned}$$

(b)

$$\begin{aligned}y &= \frac{u+1}{u-1} \\ y' &= \frac{-2}{(u-1)^2} \\ dy &= \boxed{-\frac{2}{(u-1)^2} du}\end{aligned}$$

(c)

$$\begin{aligned}y &= e^{\tan \pi t} \\ y' &= \pi \sec^2(\pi t) e^{\tan \pi t} \\ dy &= \boxed{\pi \sec^2(\pi t) e^{\tan \pi t} dt}\end{aligned}$$

(d)

$$y = \ln(1 + \sqrt{x})$$

$$y' = \frac{1}{1 + \sqrt{x}} \cdot \frac{1}{2\sqrt{x}}$$

$$dy = \boxed{\frac{1}{2\sqrt{x} + 2x} dx}$$

(e)

$$y = \arctan \frac{1}{x}$$

$$y' = \frac{1}{1 + \frac{1}{x^2}} \cdot \frac{-1}{x^2}$$

$$dy = \boxed{-\frac{1}{x^2 + 1} dx}$$

(f)

$$d(\arctan x) = \frac{1}{1 + x^2} dx$$

$$d(\arctan \frac{1}{x}) + d(\arctan x) = 0$$

$\arctan x + \arctan \frac{1}{x}$  is constant on both  $(0, \infty)$  and  $(-\infty, 0)$ .

(g)

$$y = e^{\frac{x}{10}}$$

$$y' = \frac{1}{10} e^{\frac{x}{10}} \implies y'(0) = \frac{1}{10}$$

$$dy = \frac{1}{10} \cdot 0.1 = 0.01$$

$$\Delta y = e^{0.01} - 1 \approx 0.010005$$

$$|\Delta y - dy| = 0.000005$$

$$y = \tan x$$

$$y' = \sec^2 x \implies y'(\frac{\pi}{4}) = 2$$

$$dy = 2 \cdot -0.1 = -0.2$$

$$\Delta y = \tan(\frac{\pi}{4} - 0.1) - 1 \approx -0.18237$$

$$|\Delta y - dy| = 0.01763$$

The first approximation is much better, as its second derivative's magnitude is closer to zero.

(h)

$$\begin{aligned}
 x^2 + xy + y^2 &= 7 \\
 2x \, dx + x \, dy + y \, dx + 2y \, dy &= 0 \\
 dy &= -\frac{2x+y}{2y+x} \, dx = -\frac{4}{5} \, dx \\
 y &= -\frac{4}{5}(x-1) + 2 = 1.92 \\
 y &= 1.918
 \end{aligned}$$

**Problem 4**

(a)

$$\begin{aligned}
 f(-1) &= 4 \\
 f'(-1) &= -4 - 6 = -10 \\
 y - 4 &= -10(x + 1)
 \end{aligned}$$

(b)

$$\begin{aligned}
 f\left(\frac{\pi}{2}\right) &= 0 \\
 f'\left(\frac{\pi}{2}\right) &= -1 \\
 y &= -\left(x - \frac{\pi}{2}\right)
 \end{aligned}$$

(c)

$$\begin{aligned}
 f(1) &= 0 \\
 f'(1) &= 1 \\
 y &= x - 1
 \end{aligned}$$

(d)

$$\begin{aligned}
 f(0) &= 1 \\
 f'(0) &= k \\
 y - 1 &= kx
 \end{aligned}$$

**Problem 4**

(a)

$$L(x) = 2 + \frac{1}{4}(x - 1)$$

(b)

$$\begin{aligned}
 f'(x) &= \frac{1}{2\sqrt{x+3}} \\
 f''(x) &= -\frac{1}{4\sqrt{(x+3)^3}}
 \end{aligned}$$

$f(x)$  is concave down on its domain. Hence, both approximations are overestimates.

**Problem 5**

(a)

$$f(x) = x^{\frac{2}{3}}$$

$$f'(x) = \frac{2}{3\sqrt[3]{x}}$$

$$L(x) = \frac{1}{3}(x - 8) + 4$$

$$L(8.06) = \boxed{4.02}$$

**Problem 6**

(b)

$$r = 30 \pm 0.1$$

$$V = r^3$$

$$dV = 3r^2 dr = 3(30)^2 0.1 = \boxed{270}$$

$$\frac{dV}{V} = \frac{270}{30^3} = \boxed{\frac{1}{100}}$$

$$A = 6r^2$$

$$dA = 12r dr = 12(30)0.1 = \boxed{36}$$

$$\frac{dA}{A} = \frac{36}{6 \cdot 30^2} = \boxed{\frac{1}{150}}$$

(c)

$$Q = kx^n$$

$$dQ = knx^{n-1} dx$$

$$\boxed{\frac{dQ}{Q} = n \frac{dx}{x}}$$

The relative error is for a quantity proportional to the  $n$ -th power of another variable is equal to  $n$  times the relative error for that variable.

**Problem 7**

(a)

$$L(x) = 2(x - 1) + 5 = 2x + 3$$

$$f(0.9) \approx \boxed{4.8}$$

$$f(1.1) \approx \boxed{5.2}$$

(b) As  $f'(x)$  is decreasing on its interval,  $f''(x) < 0$ . Hence, the approximations are overestimates.

(c)  $f(1) - f(0.9)$  is larger, as the area under  $f'(x)$  for  $(0.9, 1)$  is greater than the area for  $(1, 1.1)$ .

(d)

$$L(3) = 9$$

$$f(3) \approx 8$$

I would trust the trapezoidal estimate more, as the slope changes significantly from 1 to 3 and the linearization method is simply not precise.

## Problem 8

(a)

$$h = 50 \tan \frac{\pi}{3} = \boxed{50\sqrt{3}}$$

$$h = 50 \tan \theta$$

$$dh = 50 \sec^2 \theta d\theta$$

$$dh = 50 \cdot 4 \cdot \frac{\pi}{360} = \boxed{\frac{5\pi}{9}}$$

$$\frac{dh}{h} = \frac{5\pi}{9} \cdot \frac{1}{50\sqrt{3}} = \boxed{\frac{\pi}{90\sqrt{3}}}$$

(b)

$$h = d \tan \theta$$

$$dh = d \sec^2 \theta d\theta$$

$$\frac{dh}{h} = \frac{\sec^2 \theta}{\tan \theta} = \frac{d\theta}{\sin \theta \cos \theta} = \boxed{\frac{2 d\theta}{\sin 2\theta}}$$

(c)

$$\sin 2\theta = 1 \implies \theta = \boxed{\frac{\pi}{4}}$$

$$d = h$$

$$|d\theta| = \frac{\pi}{360} \implies E = \boxed{\frac{\pi}{180}}$$

(d) When standing either very far away or standing very close to the object, any slight difference in  $\theta$  may lead to huge differences in the calculated value.

## Problem 9

(a)

$$(1+x)^k \approx 1+kx$$

$$k = \frac{1}{3} \implies (1-x)^{\frac{1}{3}} \approx 1 - \frac{1}{3}x$$

$$k = -4 \implies (1+2x)^{-4} \approx 1 - 8x$$

(b)

$$\begin{aligned}
|(1-x)^{\frac{1}{3}} - (1 - \frac{1}{3}x)| &\leq 0.1 \\
\iff -1.20 &\leq x \leq 0.71 \\
|(1+2x)^{-4} - (1-8x)| &\leq 0.1 \\
\iff -0.05 &\leq x \leq 0.06
\end{aligned}$$

The magnitude of the second function's second derivative is much larger than the first function's near  $a = 0$ .

(c)

$$K = mc^2 \left( \frac{1}{\sqrt{1-v^2/c^2}} - 1 \right) = mc^2 \left( 1 + \frac{v^2}{2c^2} - 1 \right) = \boxed{\frac{1}{2}mv^2}$$

It is an understatement, as the second derivative of the function  $f(x) = \frac{1}{\sqrt{1-x}}$  is greater than zero.

(d)

$$\begin{aligned}
\ln(1) &= 0 \\
\frac{d}{dx} \ln(1) &= 1 \\
y &= x \\
T &= \frac{\ln 2}{\ln(1+R/100)} = \frac{100 \ln 2}{R} \approx \boxed{\frac{69.3}{R}}
\end{aligned}$$

It is an underestimate as the second derivative of the linearization function is greater than zero near  $R = 0$ .

## Problem 10

(a)

$$\begin{aligned}
L(x) &= f(x_n) + f'(x_n)(x - x_n) \\
L(x_{n+1}) &= 0 \\
f(x_n) + f'(x_n)(x_{n+1} - x_n) &= 0 \\
\boxed{x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)}}
\end{aligned}$$

(b)

$$\begin{aligned}
f(x) &= x^2 - 2 \\
f'(x) &= 2x \\
x_{n+1} &= x_n - \frac{x_n^2 - 2}{2x_n} = \frac{1}{2} \left( x_n + \frac{2}{x_n} \right)
\end{aligned}$$

$$\begin{aligned}
 x_0 &= 1 \\
 x_1 &= \frac{3}{2} = 1.5 \\
 x_2 &= \frac{17}{12} = 1.41\bar{6} \\
 x_3 &= \frac{577}{408} = 1.4142
 \end{aligned}$$

(c)

$$\begin{aligned}
 x_{n+1} &= \frac{x_n^2 + 2}{2x_n} \\
 x_{n+1} - \sqrt{2} &= \frac{x_n^2 + 2}{2x_n} - \sqrt{2} = \frac{x_n^2 + 2\sqrt{2}x_n + 2}{2x_n} = \frac{(x_n - \sqrt{2})^2}{2x_n}
 \end{aligned}$$

The absolute error is squared now, rendering the number of correct decimal places to also double.

(d)

$$\begin{aligned}
 x_{n+1} &= x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2} \\
 x_0 &= 2 \\
 x_1 &= 2.1 \\
 x_2 &= 2.09456 \\
 x_3 &= 2.09455
 \end{aligned}$$

(e)

$$\begin{aligned}
 x_{n+1} &= -2x_n \\
 x_0 &= 1 \\
 x_1 &= -2 \\
 x_2 &= 4 \\
 x_3 &= -8
 \end{aligned}$$

The sequence oscillates as the derivative of the function approaches infinity as  $x$  approaches 0.

### Problem 11

(a)

$$\begin{aligned}
 f\left(\frac{1}{3}\right) &= \frac{3}{10} \\
 f(x) &= \frac{x}{1+x^2}
 \end{aligned}$$

$$\begin{aligned}
 f'(x) &= \frac{1-x^2}{(1+x^2)^2} \\
 f\left(\frac{1}{3}\right) &= \frac{\frac{8}{9}}{\frac{100}{81}} = \frac{18}{25} \\
 L(x) &= \frac{3}{10} + \frac{18}{25}\left(x - \frac{1}{3}\right) = \frac{18}{25}x + \frac{3}{50} \\
 f(x) &\leq L(x) \\
 \frac{x}{1+x^2} &\leq \frac{18}{25}x + \frac{3}{50} \\
 50x &\leq (1+x^2)(36x+3) \\
 36x^3 + 3x^2 - 14x + 3 &\geq 0 \\
 (3x-1)^2(4x+3) &\geq 0
 \end{aligned}$$

(b)

$$\begin{aligned}
 \frac{a}{a^2+1} + \frac{b}{b^2+1} + \frac{c}{c^2+1} &\leq \frac{18}{25}(a+b+c) + \frac{9}{50} = \frac{18}{25} + \frac{9}{50} = \frac{9}{10} \\
 \boxed{\frac{a}{a^2+1} + \frac{b}{b^2+1} + \frac{c}{c^2+1} &\leq \frac{9}{10}} \\
 \boxed{a=b=c=\frac{1}{3}}
 \end{aligned}$$

(c)

$$\begin{aligned}
 B(p) &= 3f(p) + f'(p)(1-3p) \\
 B'(p) &= 3f'(p) + f''(1-3p) - 3f'(p) = f''(1-3p) \\
 f''(p) &= \frac{d}{dp} \frac{1-p^2}{(1+p^2)^2} = \frac{2x(x^2-3)}{(1+p^2)^3} \\
 B'(p) &= \frac{2x(x^2-3)(1-3p)}{(1+p^2)^3} \\
 B'(p) = 0 &\implies \boxed{p = \frac{1}{3}}
 \end{aligned}$$

(d)

$$\begin{aligned}
 b+c &= 3-a \\
 \frac{1}{a^2+b+c} &= \frac{1}{a^2-a+3} \\
 f(x) &= \frac{1}{x^2-x+3} \\
 f(1) &= \frac{1}{3} \\
 f'(1) &= -\frac{1}{9}
 \end{aligned}$$

$$L(x) = -\frac{1}{9}(x-1) + \frac{1}{3} = \frac{4-x}{9}$$

$$f(x) \leq L(x), \quad x \in (0, 3)$$

$$\frac{1}{a^2+b+c} + \frac{1}{b^2+c+a} + \frac{1}{c^2+a+b} \leq \frac{12-(a+b+c)}{9} = 1$$

$$\boxed{\frac{1}{a^2+b+c} + \frac{1}{b^2+c+a} + \frac{1}{c^2+a+b} \leq 1}$$